

Exp no:8 -Determination and plotting of Capacitance of Co-axial cable per unit length with different (i)inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Aim:To write program for Capacitance of co-axial cable per unit length using SCILAB.

Software required:

- SCILAB version 6.0.1



The screenshot displays the SCILAB 2026.1.0 environment. The top window is the 'SciNotes' editor, showing a script named 'emf8.0.sci'. The script calculates the capacitance per unit length of a coaxial cable based on user input for inner radius (r1), outer radius (r2), and relative permittivity (er). It includes a validation check to ensure r2 is greater than r1. The script uses the formula $C = \frac{2 \cdot \pi \cdot \epsilon \cdot \epsilon_r}{\log(r2 / r1)}$ to calculate the capacitance (C) in F/m. The bottom window is the 'Scilab 2026.1.0 Console', which shows the execution of the script. It prompts the user to enter values for r1, r2, and er, and then displays the calculated capacitance per unit length of the coaxial cable.

```
1 clc;
2 clear;
3
4 //Input-values
5 r1 = input("Enter the inner radius r1 (m) : ");
6 r2 = input("Enter the outer radius r2 (m) : ");
7 er = input("Enter the relative permittivity (er) : ");
8
9 //Constant
10 e0 = 8.854e-12;
11 e = e0 * er;
12
13 //Check-validity
14 if r2 <= r1 then
15     disp("Error:-r2 must be greater than r1.");
16 else
17     //Capacitance-per-unit-length
18     c = (2 * %pi * e) ./ log(r2 / r1);
19
20     disp("Capacitance per unit length of coaxial cable (F/m) :");
21     disp(c);
22
23     //Plot
24     r_plot = linspace(r1, r2*5, 50);
25     c_plot = (2 * %pi * e) ./ log(r_plot ./ r1);
26
27     clf(0);
28     plot2d(r_plot, c_plot);
29     xtitle("Capacitance of Coaxial Cable", ...
30           "Outer Radius r2 (m)", ...
31           "Capacitance (F/m)");
32     xgrid();
33 end
```

Scilab 2026.1.0 Console

```
File Edit Control Applications ?
[Icons]
Scilab 2026.1.0 Console

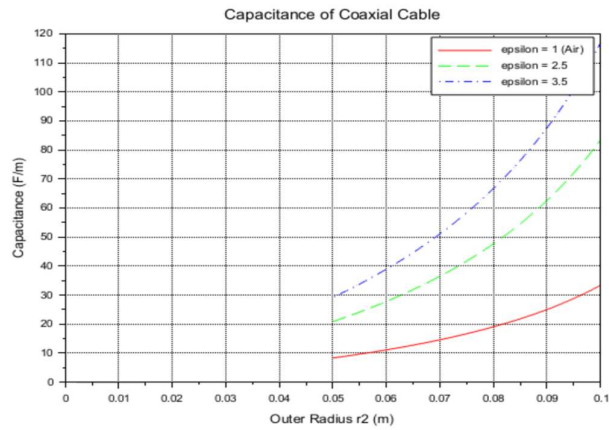
Enter the inner radius r1 (m) : 2*10^-3

Enter the outer radius r2 (m) : 3*10^-3

Enter the relative permittivity (er) : 5

"Capacitance per unit length of coaxial cable (F/m) : "
6.860D-10

--> |
```



⑧ Given:-

$r_1 = 2 \text{ mm}$
 $r_2 = 3 \text{ mm}$
 $\epsilon_r = 5$

Formula:-

$$C = \frac{2\pi\epsilon}{\ln\left(\frac{r_2}{r_1}\right)}$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\epsilon = (8.854 \times 10^{-12}) \times 5$$

$$\epsilon = 4.427 \times 10^{-11} \text{ F/m}$$

$$\frac{r_2}{r_1} = \frac{0.003}{0.002} = 1.5$$

$$\ln(1.5) = 0.4055$$

$$2\pi\epsilon = 2 \times 3.1416 \times 4.427 \times 10^{-11}$$

$$= 2.7815 \times 10^{-10}$$

$$C = \frac{2.7815 \times 10^{-10}}{0.4055}$$

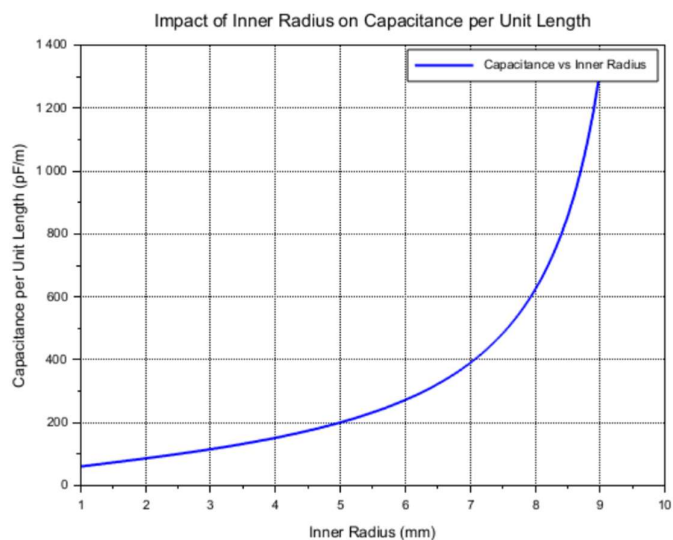
$$C = 6.860 \times 10^{-10} \text{ F/m}$$

$C = 6.860 \times 10^{-10} \text{ F/m}$

CASE STUDY 1: Variation with Inner Radius

```
8.1.sci (C:\Users\abhis\Documents\8.1.sci) - SciNotes
File Edit Format Options Window Execute ?
8.1.sci (C:\Users\abhis\Documents\8.1.sci) - SciNotes
kihuy.sci x EMF4.1.sci x EMF4.2.sci x emf4.3.sci x Untitled 1 x em

3 clear;
4 clf;
5
6 //Constants
7 epsilon_0 = 8.854e-12; //Permittivity of free space (F/m)
8 epsilon_r = 2.5; //Relative permittivity of dielectric
9
10 //Outer radius
11 r2 = 0.01; //1 cm = 0.01 m
12
13 //Inner radius range
14 r1_min = 0.001; //0.1 cm = 1 mm
15 r1_max = r2 - 0.001; //9 mm
16
17 //Generate inner radius values
18 r1 = r1_min:0.0001:r1_max;
19
20 //Capacitance per unit length
21 //C = 2*pi*epsilon./ln(r2/r1)
22 C = (2.*pi.*epsilon_r.*epsilon_0)./log(r2./r1);
23
24 //Plot
25 plot(r1.*1000, C.*1e12, 'b-', 'LineWidth', 2);
26
27 //Labels
28 xlabel('Inner Radius (mm)');
29 ylabel('Capacitance per Unit Length (pF/m)');
30 title('Impact of Inner Radius on Capacitance per Unit Length');
31
32 //Grid
33 xgrid();
34
35 //Legend
36 legend('Capacitance vs Inner Radius', 'location', 'northwest');
37
38 //Display message
39 disp('Plot generated: Capacitance vs Inner Radius');
40
```



CASE STUDY 2: Variation with Outer Radius

```

8.2.sci (C:\Users\abhis\Documents\8.2.sci) - SciNotes
File Edit Format Options Window Execute ?

8.2.sci (C:\Users\abhis\Documents\8.2.sci) - SciNotes
khuy.sci x EMF4.1.sci x EMF4.2.sci x emf4.3.sci x Untitled 1 x emf7.0

7 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
8 epsilon_r = 3.0; // Relative permittivity
9 a = 1e-3; // Inner radius = 1 mm
10
11 // Outer radius values from 5 mm to 20 mm
12 b_values = 5e-2:1e-2:20e-2;
13
14 // Initialize array for capacitance per unit length
15 C_prime = zeros(size(b_values));
16
17 // Loop through outer radius values
18 for i = 1:length(b_values)
19     b = b_values(i);
20     // Permittivity of dielectric
21     epsilon = epsilon0 * epsilon_r;
22     // Capacitance per unit length
23     C_prime(i) = (2 * pi * epsilon) / log(b / a);
24 end
25
26 // Plotting
27 axis(0);
28 plot(b_values * 1e3, C_prime * 1e12, 'b-o');
29
30 // Labels
31 xlabel('Outer Radius (mm)');
32 ylabel('Capacitance per Unit Length (pF/m)');
33 title('Variation of Capacitance per Unit Length with Outer Radius');
34
35 // Grid
36 grid();
37
38 // Display results
39 disp('Outer Radius (mm) and Capacitance per Unit Length (pF/m):');
40 disp([b_values * 1e3, C_prime * 1e12]);
41
42

```



```

SciLab 2026.1.0 Console
File Edit Control Applications ?

SciLab 2026.1.0 Console

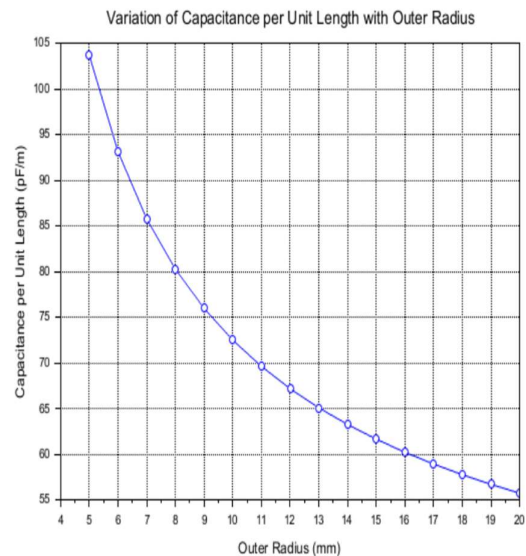
```

```

"Outer Radius (mm) and Capacitance per Unit Length (pF/m):"
5. 103.69705
6. 93.145297
7. 85.766534
8. 80.259033
9. 75.956718
10. 72.481129
11. 69.600191
12. 67.163074
13. 65.067159
14. 63.239993
15. 61.628831
16. 60.194275
17. 58.906248
18. 57.74135
19. 56.681076
20. 55.710575

```

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CASE STUDY 3: Variation with Relative Permittivity

```

8,3.sci (C:\Users\abhis\Documents\8,3.sci) - SciNotes
File Edit Format Options Window Execute ?
8,3.sci (C:\Users\abhis\Documents\8,3.sci) - SciNotes
khuy.sc EMF4.1.sc EMF4.2.sc emf4.3.sc Untitled 1 emf7.0.sc

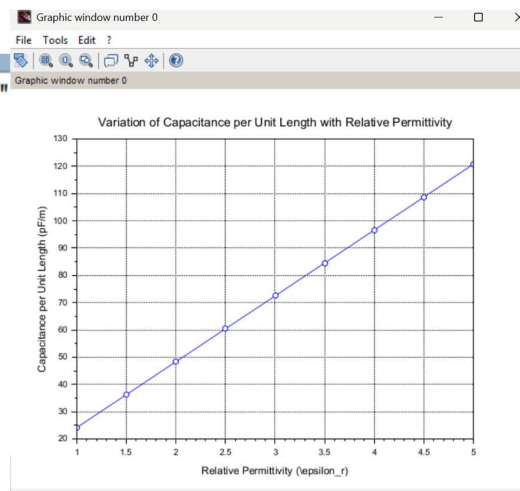
1 // epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
2 a = 1e-3; // Inner radius = 1 mm
3 b = 10e-3; // Outer radius = 10 mm
4 // Relative permittivity values from 1 to 5
5 epsilon_r_values = 1:0.5:5;
6 // Initialize array for capacitance per unit length
7 C_prime = zeros(size(epsilon_r_values));
8 // Calculate capacitance for each relative permittivity
9 for i = 1:length(epsilon_r_values)
10     epsilon_r = epsilon_r_values(i);
11     // Permittivity of dielectric
12     epsilon = epsilon0 * epsilon_r;
13     // Capacitance per unit length
14     C_prime(i) = (2 * pi * epsilon) / log(b / a);
15 end
16 // Plotting
17 clf(0);
18 plot(epsilon_r_values, C_prime * 1e12, 'b-o');
19 // Labels
20 xlabel('Relative Permittivity (\epsilon_r)');
21 ylabel('Capacitance per Unit Length (pF/m)');
22 title('Variation of Capacitance per Unit Length with Relative Permittivity');
23 // Grid
24 grid();
25 // Display results
26 disp('Relative Permittivity and Capacitance per Unit Length:');
27 disp([epsilon_r_values' C_prime' * 1e12]);
28

```

```

SCILAB 2020-10 CONSOLE
"Relative Permittivity and Capacitance per Unit Length:"
1.    24.160376
1.5   36.240565
2.    48.320753
2.5   60.400941
3.    72.481129
3.5   84.561318
4.    96.641506
4.5   108.72169
5.    120.80188
-->

```



Result : Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.